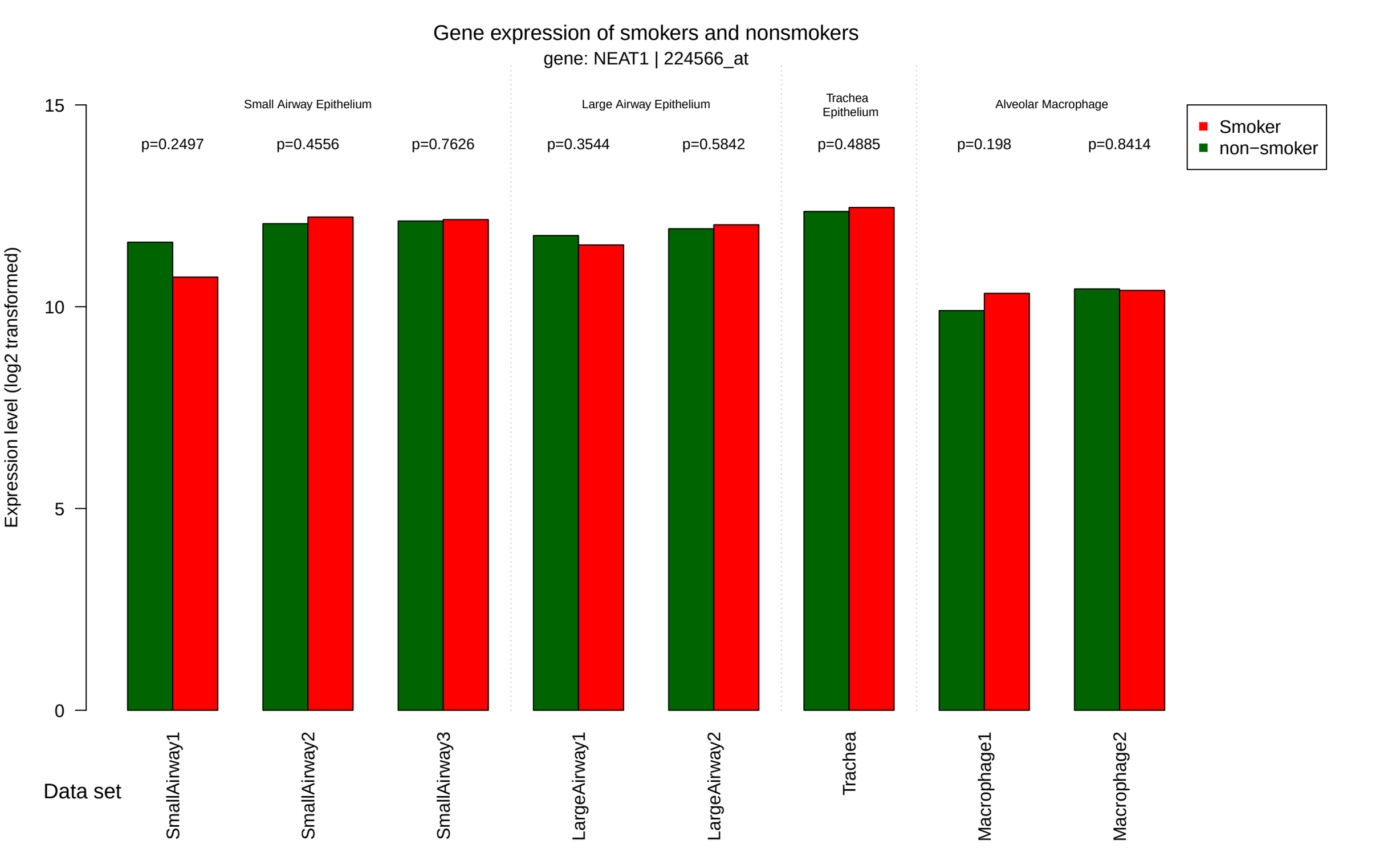
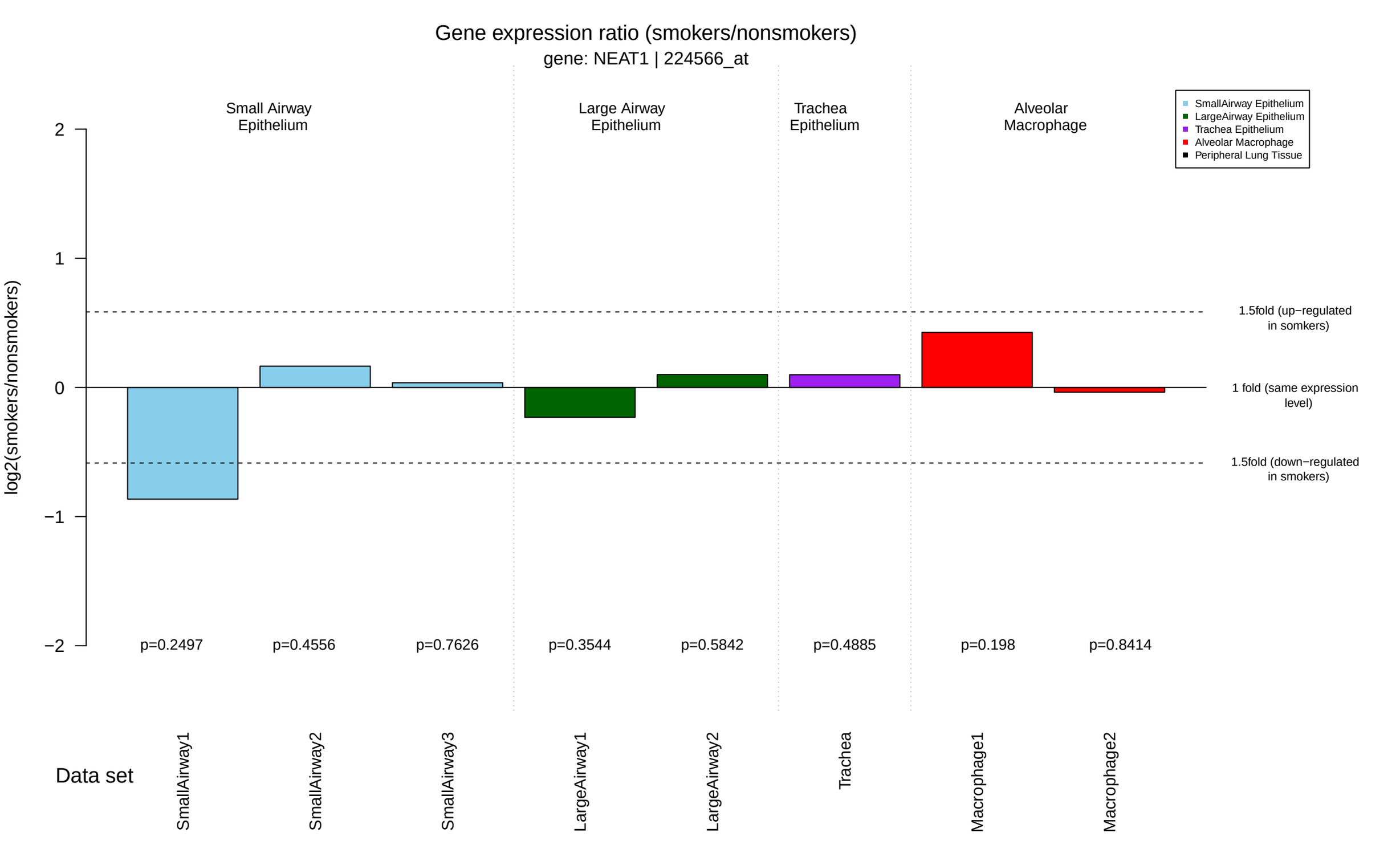
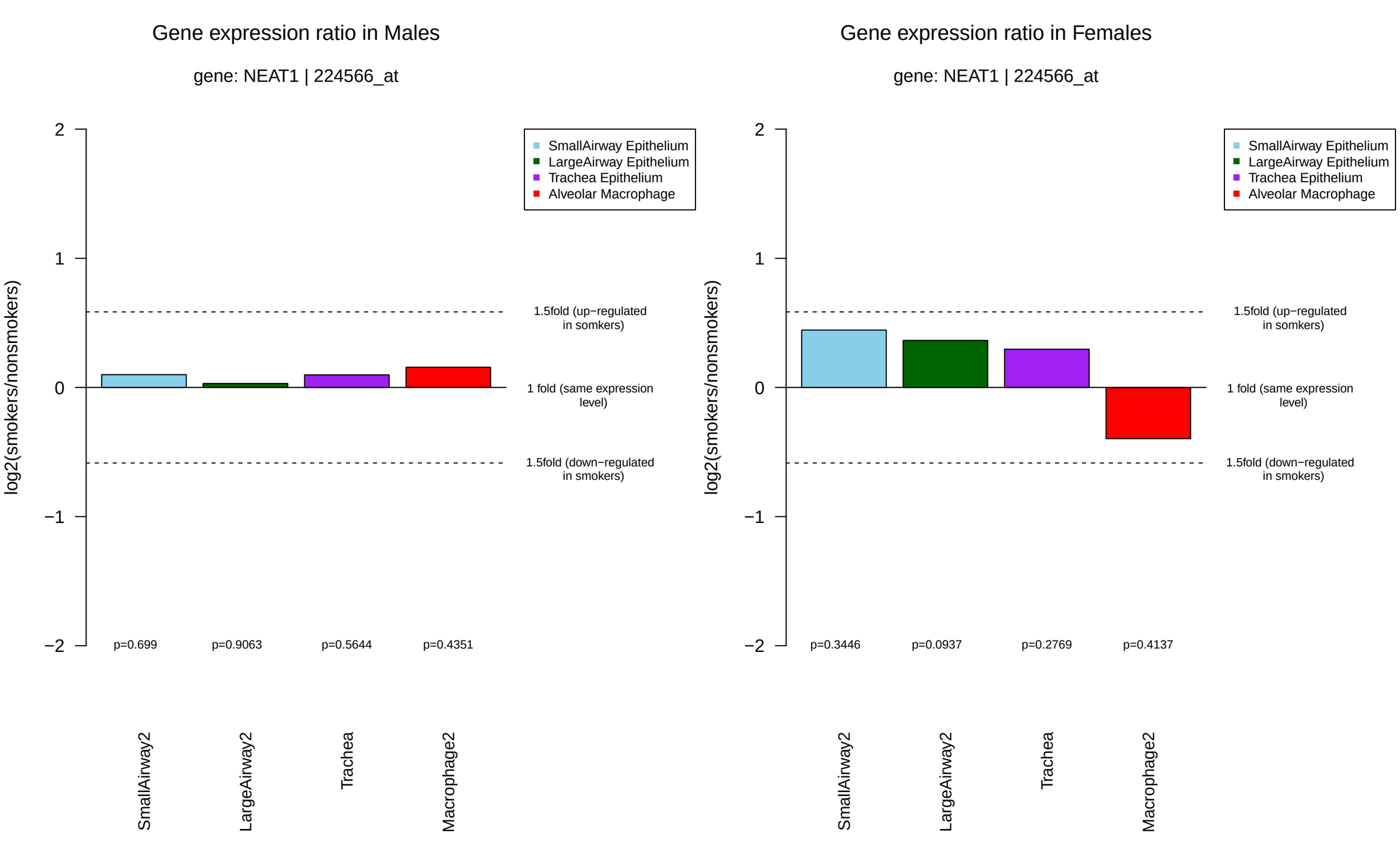
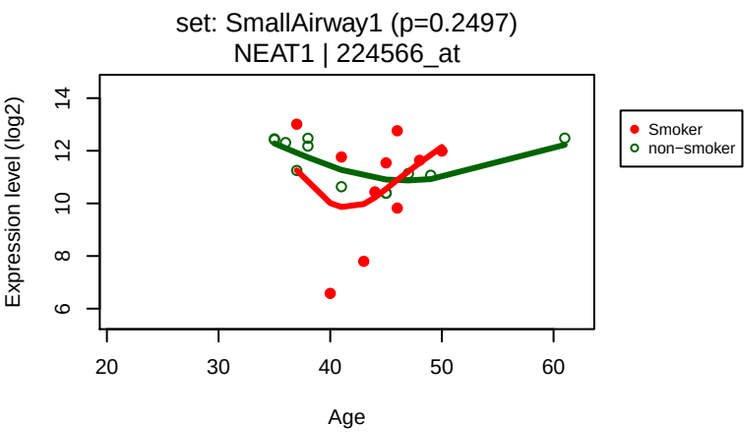
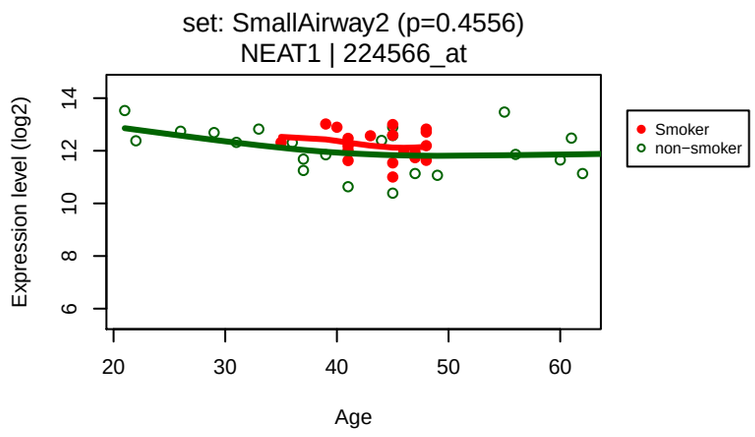
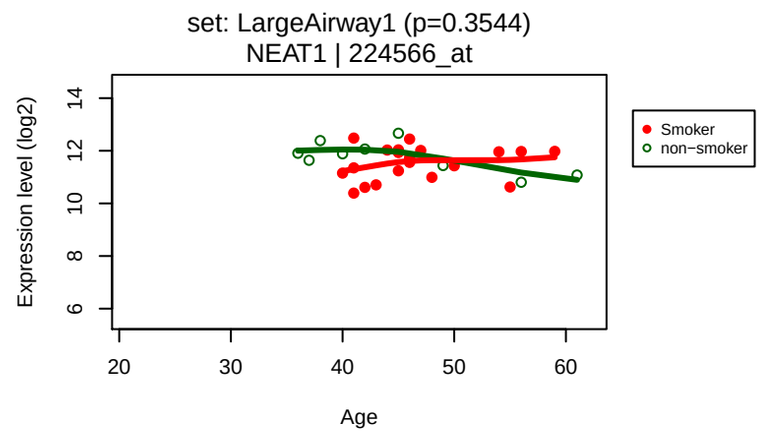

Gene expression of smokers and nonsmokers
gene: NEAT1 | 224566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2497
p=0.4556
p=0.7626
p=0.3544
p=0.5842
p=0.4885
p=0.198
p=0.8414
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

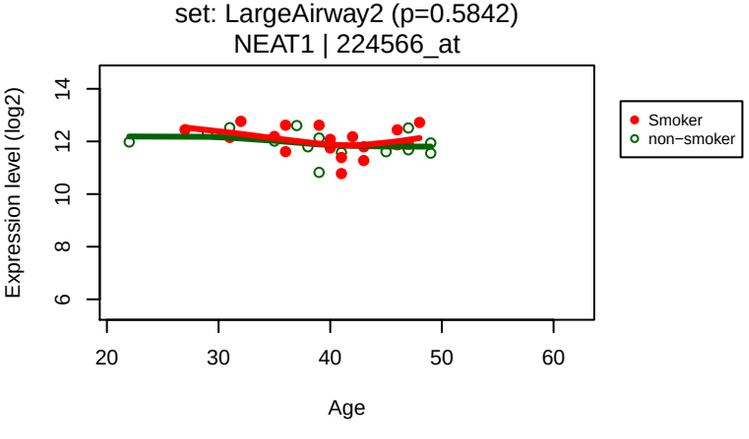

Gene expression ratio (smokers/nonsmokers)
gene: NEAT1 | 224566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2497
p=0.4556
p=0.7626
p=0.3544
p=0.5842
p=0.4885
p=0.198
p=0.8414
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

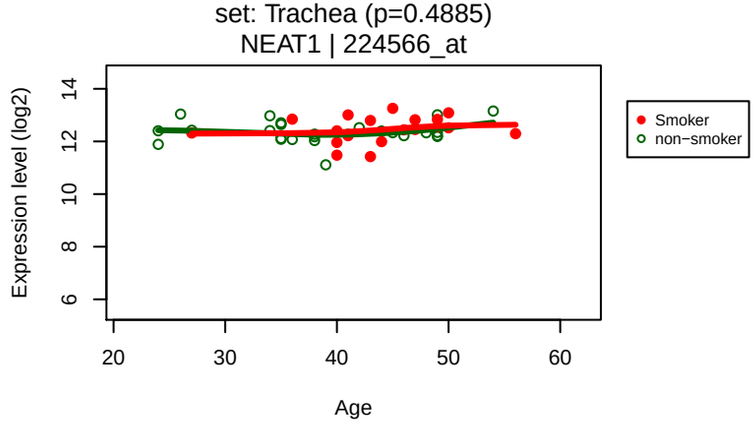

Gene expression ratio in Males
gene: NEAT1 | 224566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.699
p=0.9063
p=0.5644
p=0.4351
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEAT1 | 224566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3446
p=0.0937
p=0.2769
p=0.4137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

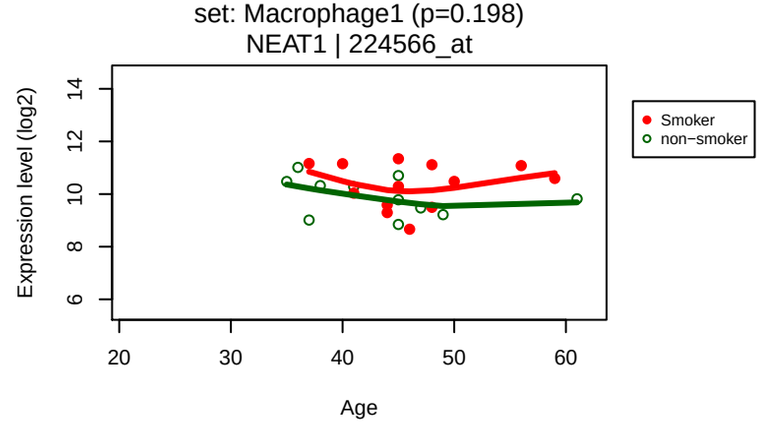

set: SmallAirway1 (p=0.2497)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker

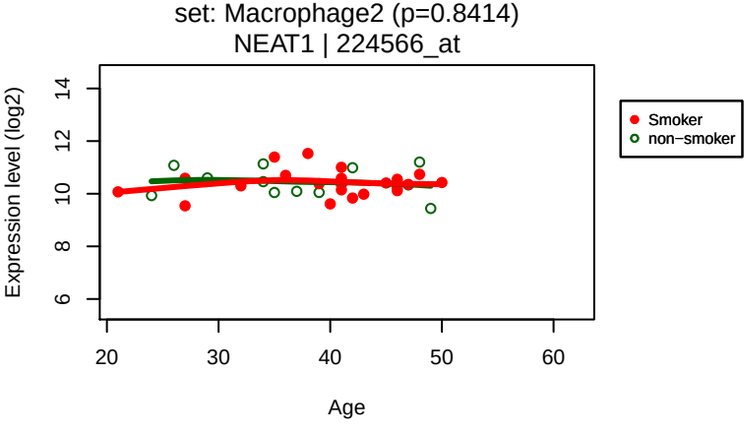

set: SmallAirway2 (p=0.4556)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker

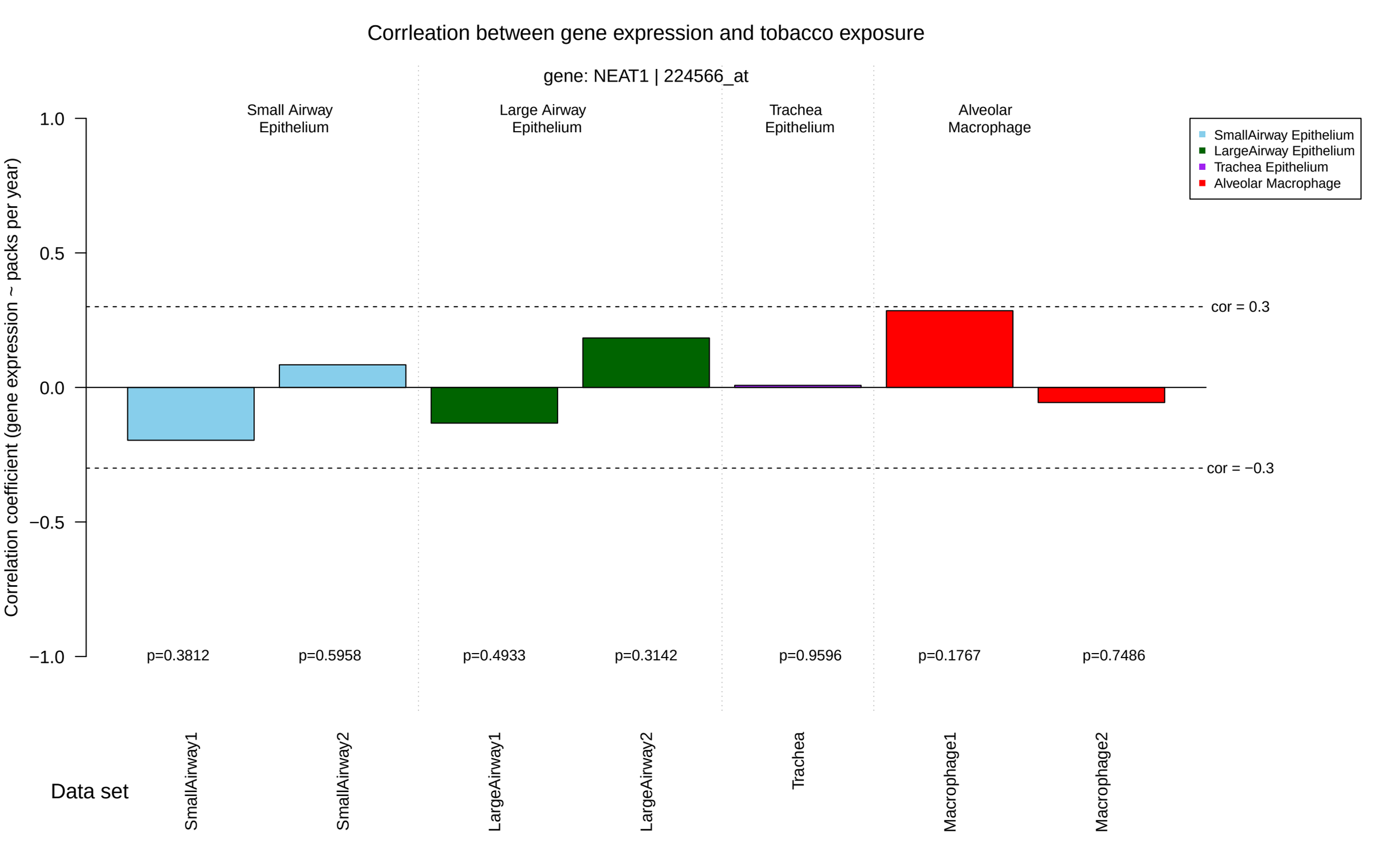

set: LargeAirway1 (p=0.3544)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5842)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4885)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.198)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8414)
NEAT1 | 224566_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEAT1 | 224566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3812
p=0.5958
p=0.4933
p=0.3142
p=0.9596
p=0.1767
p=0.7486
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2